InfoServer

A GPS-Disciplined Stratum-1 Network Time and Information Server

Operator Manual

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This manual covers the setup, configuration, and day-to-day operation of the InfoServer — a self-contained network time server that derives time from a GPS receiver (or an upstream Internet time server), disciplines a temperature-compensated DS3231 real-time clock to that source, and serves it to NTP clients on the local network. It also publishes local weather and sensor readings on the network and can send alerts by email or Telegram message on time-source faults. It includes a Theory of Operation chapter explaining how the server produces and distributes accurate time.

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Quick Start

The InfoServer is a stratum-1 network time server. It keeps highly accurate time from a GPS receiver (or, as a fallback, from an Internet time server) and hands that time out to every computer, clock, and device on the local network using the standard NTP protocol. It can also publish local weather and environmental readings on the network, and send an email or Telegram alert if a time-source problem occurs. A full explanation of every option is in the numbered sections that follow.

1. Position the server and power it

The GPS antenna is built into the receiver, so there is nothing to attach — just place the server itself where it can see the sky, such as a window sill or, ideally, outdoors. Connect a 5 V USB-C power supply. On a brand-new unit, AP setup mode starts automatically (see step 2).

2. Connect to the server's WiFi

From a phone or computer, open WiFi settings and look for the network named after the server — InfoServer by default. Connect to it; no password is required for the setup network.

3. Configure in a browser

Open a web browser and go to 192.168.4.1. The setup page loads. At minimum:

- Select the home WiFi network from the SSID dropdown and enter its password.
- Leave the Discipline Mode at NMEA GPS unless the server is to run from an Internet time server instead of GPS.

Tap Save, then Start. The server saves the settings and reboots onto the home network.

4. Wait for a GPS fix

After reboot the server joins the home WiFi network and begins serving as soon as a time source is locked. A first GPS fix from a cold start can take from under a minute to several minutes depending on how much open sky the unit can see. The STATUS LED shows progress (Section 10), and the status web page shows the live state.

5. Point devices at it

On each device, set the NTP server address to this unit's IP address (shown on the status page) or its mDNS name — infoserver.local by default, or <name>.local if the Access Point / mDNS Name is changed. The server answers standard NTP requests on UDP port 123 — no client software is required beyond the NTP client already built into every operating system.

6. Daily use

To do this	Do this
See live status	Browse to the server's IP address (or http://infoserver.local/) from any device on the same WiFi
Read weather / sensor data	Fetch http://infoserver.local/info from any device (see Section 7)
Get email or Telegram alerts on time faults	Enable and configure them on the Notifications page (see Section 9)
Show the IP address on the LEDs	Double-click SW1 — the IP blinks out octet by octet
Install new software over WiFi	Hold SW1 for 10 seconds, or use Update Firmware on the System sub-page
Let updates install themselves	Leave Automatic Updates on (the default) — see Section 12
Add the page to a phone's home screen	Tap Add to Home Screen on the System sub-page
Re-enter setup (AP) mode	Hold SW1 for 5 seconds, or use Switch to AP Mode on the status page (System sub-page)
Factory reset	Hold SW1 while applying power; the unit wipes its settings and reboots into AP mode

1. Overview

The InfoServer is a self-contained network time and information server built on an ESP32-S3. It derives time from one of two sources — a u-blox GPS receiver or an upstream Internet NTP server — disciplines a temperature-compensated DS3231 real-time clock to that source, and serves the result to NTP clients on the local network. Alongside the time service it can publish environmental and weather information on the LAN as simple JSON, and send an alert by email or Telegram message when a time-source problem arises. A web interface handles all configuration; there is no app to install and no cloud account.

Why run a local time server?

- Accuracy. A GPS-disciplined server delivers time within roughly a millisecond of true UTC on the LAN — far tighter than typical Internet sync, and with no dependence on the Internet connection once GPS is locked.
- Independence. Devices sync to a source on the local network. They stay in agreement with each other even if the Internet is down.
- Privacy and control. NTP requests stay on the LAN rather than going out to public time servers.
- More than time. The same box can serve local weather and sensor readings to local devices and dashboards, and send an alert by email or Telegram if timekeeping is disrupted.

What's in the box

- The InfoServer unit (ESP32-S3 controller, DS3231 RTC with battery backup, an optional environmental sensor, indicator and under-light LEDs).
- A u-blox GPS receiver with a built-in antenna and a 1PPS (one-pulse-per-second) output.
- A USB-C port for power — use any standard 5 V USB-C supply (power supply not included).

Key features

- Stratum-1 GPS discipline using NMEA time plus the 1PPS edge for sub-second accuracy.
- Automatic NTP fallback if GPS is unavailable, with a configurable holdover window.

- Battery-backed DS3231 RTC carries time through brief power and signal interruptions.
- Local information services: a JSON /info manifest publishing on-board sensor readings with a barometric trend, current weather and forecast from up to three providers (OpenMeteo needs no API key), active National Weather Service alerts, and locally computed sunrise/sunset and moon-phase data (Section 7).
- Email and Telegram notifications on time-source faults, with built-in test buttons (Section 9).
- Web-based setup and a live status page; mDNS discovery as infoserver.local by default (follows the Access Point / mDNS Name).
- Over-the-air software updates, either on demand or installed automatically overnight — settings and WiFi credentials are always preserved (Section 12).
- The web interface can be added to a phone or tablet home screen and opened like an app.
- Two status indicators plus two color-configurable under-lights.

2. Theory of Operation

This section explains how the InfoServer produces and distributes accurate time. It is background reading — none of it is required to use the server, but it explains what the status page is reporting and why the design behaves as it does.

The problem NTP solves

Every computer has a clock, but those clocks drift — a few seconds a day is typical. The Network Time Protocol (NTP) lets a device ask a time server "what time is it?", measure how long the question-and-answer round trip took, and correct its own clock accordingly. The key trick is that NTP timestamps the request on the way out and the reply on the way back, so the client can subtract the network delay and recover the true time the server held at a known instant. The quality of everyone's clock therefore depends on the quality of the server's clock — which is what this device is built to maximize.

Stratum: the time hierarchy

NTP arranges time sources in numbered tiers called strata. Stratum 0 is a reference clock itself — an atomic clock or, here, the GPS system, whose satellites carry atomic clocks. A server directly attached to a stratum-0 reference is stratum 1. A server that syncs from a stratum-1 server is stratum 2, and so on; each hop adds a little uncertainty. When this unit is GPS-disciplined it advertises stratum 1. When it falls back to an Internet NTP server (typically stratum 2) it advertises stratum 3.

The time source: a disciplined RTC

At the heart of the server is a DS3231 — a temperature-compensated real-time clock chip accurate to roughly two parts per million (about a tenth of a second per day on its own). The DS3231 is not trusted to keep perfect time by itself; instead it is continuously corrected — "disciplined" — by a more accurate source. Between corrections, the server does not read the chip's ticking registers for the served time. It keeps the corrected moment in memory and extrapolates forward using the processor's own millisecond timer, which yields a smooth, phase-accurate clock that can report not just the second but the precise fraction of a second.

Discipline source 1: GPS

In the default mode the discipline source is the GPS receiver. GPS is two signals working together. First, the receiver sends NMEA sentences over a serial line —

ordinary text messages that include the current date and time to the whole second. Second, and crucially, it emits a 1PPS pulse: a sharp electrical edge that occurs exactly on each UTC second boundary, accurate to tens of nanoseconds. The NMEA text says which second it is; the 1PPS edge says exactly when that second begins. The firmware combines the two — taking the whole second from the text and the precise instant from the pulse — to anchor the served clock to true UTC within well under a millisecond. This is what makes the server stratum 1.

Discipline source 2: upstream NTP

When running without GPS, or if GPS never acquires a fix, the server can instead discipline itself from an Internet NTP server (for example pool.ntp.org or time.google.com). This is less accurate — Internet round trips introduce some tens of milliseconds of uncertainty — but it needs no view of the sky and still produces a perfectly usable LAN time server. In this mode the unit advertises stratum 3.

Holdover: riding through a GPS dropout

GPS reception is not perfectly continuous — a passing obstruction or a poor view of the sky can briefly interrupt the fix. Because the DS3231 drifts only about seven milliseconds per hour, the server does not panic the instant GPS disappears. It enters holdover: it keeps serving GPS-derived time from the DS3231, still as stratum 1, for a configurable window (one hour to one day, default four hours). Only if GPS stays gone past that window does the server hand discipline to the NTP fallback. The logic deliberately prefers a few hours of holdover over an immediate switch, because the drifting-but-recent GPS time is actually tighter than a fresh Internet NTP sample would be.

NTP as a witness

When both GPS and an upstream NTP server are available, the NTP fallback does not sit idle. While GPS holds discipline, the server still polls the upstream NTP server on a schedule — but only to compare, not to correct. The difference between the two independent sources is shown on the status page as the GPS−NTP figure. In normal operation it is a few milliseconds, which is reassuring evidence that two unrelated time sources agree. If they ever disagree by more than a second and keep disagreeing, the figure turns red — a genuine alarm that something (a GPS fault, poor reception, a misbehaving upstream server) deserves attention.

Guarding against bad data

A time server that blindly accepted every sample could be thrown wildly off by a single garbled GPS sentence or a malformed packet. Two safeguards prevent this. First, GPS sentences are checksum-verified before use, and NTP responses are validated against the protocol. Second, a sanity filter rejects any correction that would jump the clock by more than a minute — unless the same large offset persists across several samples, in

which case it is eventually accepted as a real (if dramatic) correction. The result is a clock that tracks small, legitimate adjustments instantly but refuses to be yanked around by a one-off glitch.

Serving clients

The server listens for NTP requests on UDP port 123. When a request arrives it records the exact moment of arrival, fills in the reply with its reference identifier and stratum, stamps the precise transmit time, and sends it back. One safeguard is fundamental: the server refuses to answer at all until it has actually locked to a real time source. Before the first GPS or NTP sync it stays silent rather than handing out unverified, free-running RTC time that a client would trust. The moment a source locks, serving begins automatically.

Putting it together

On a typical boot the sequence is: the DS3231 provides a rough starting time from its backup battery (but the server does not serve yet); WiFi connects and, if configured, an NTP sample disciplines the clock within seconds; GPS then acquires a fix and takes over as the stratum-1 source, with NTP demoted to witness. From then on the served clock is anchored to the GPS 1PPS edge every second, holds over through brief GPS dropouts, and is continuously cross-checked against an independent Internet source — all while answering client requests on the LAN.

3. Hardware

The server is a single board built around an ESP32-S3 module. The major components and their roles:

Component	Role
ESP32-S3 controller	Runs the firmware: WiFi, NTP server, GPS parsing, info services, web interface, LED control
DS3231 RTC	Temperature-compensated real-time clock with coin-cell backup; the disciplined time source
u-blox GPS receiver	Supplies NMEA time messages and the 1PPS one-second pulse
Environmental sensor (optional)	A BME280 or BME68x measures temperature, barometric pressure and humidity (a BME68x adds an air-quality index); a Sensirion SEN66 measures particulates, carbon dioxide, temperature, humidity and volatile compounds. Both kinds may be fitted together. Each is identified automatically at power-on and published on the network and status page. Omitted if not fitted
4 × APA102 LEDs	Two status indicators (top) and two color-configurable under-lights (bottom)
SW1 button	IP-address readout (double-click), AP mode (5-second hold), and over-the-air update (10-second hold)
USB-C	5 V power input

A coin cell on the DS3231 preserves the time across power loss, so after a reboot the server starts from a close approximation and is back to full accuracy as soon as a source re-locks.

4. First-Time Setup (AP Mode)

A brand-new server, or one that has been factory reset, has no saved WiFi credentials and so starts in Access Point (AP) setup mode automatically. In this mode the two status LEDs alternate steadily.

Connecting

From a phone or computer, open WiFi settings and join the network named InfoServer (or whatever Access Point / mDNS Name has been configured). No password is required. Then open a browser to 192.168.4.1.

Network settings

- SSID — choose the 2.4 GHz home network from the dropdown. The ESP32 does not support 5 GHz networks. If the server already has a network saved but that network is not on the air when the page loads, it appears as the first entry marked "(saved - not in range)", so what the list shows is always what the server will actually use.
- SSID Override — optionally type a network name by hand (for a hidden network, or one not in range during setup).
- Network Password — the password for the selected network.
- Access Point / mDNS Name — up to 31 characters; the name this server advertises, used for the AP network, as its mDNS name (<name>.local), and as the hostname it reports to the router — so it appears by name, not as a generic device, in the router's client list.

Static IP Address (optional)

By default the server takes an address from the router by DHCP. Because clients are pointed at this server by address, a stable address is helpful; the most reliable approach is a DHCP reservation in the router. Alternatively, in the Static IP Address section, tick Use Static IP and fill in the IP Address, Gateway, Subnet Mask, and DNS Server — choose an address outside the router's DHCP pool. An entry that is not a valid address is refused with the message "Not a valid IP address (e.g. 192.168.4.50) - not saved".

Finishing

Tap Save to store settings without rebooting, or Start to commit and reboot onto the home network. After Start, reconnect the phone or computer to the normal WiFi network;

the server will appear at its DHCP or static address and, if the network resolves mDNS names, as infoserver.local. (Once the server is running on the network, Save persists changes live and reboots automatically only when a setting that takes effect at boot is changed — see Section 6.)

If infoserver.local does not work

The server advertises itself on the network under the name it was given, so http://infoserver.local/ is usually all that is needed. That name is resolved by mDNS, and mDNS is not supported by every device and depends heavily on the router's features and configuration. Some Windows PCs need Apple Bonjour installed, some Android versions do not resolve .local names at all, and many routers block or filter the multicast traffic mDNS relies on — particularly on guest networks, or with client isolation, or when 2.4 and 5 GHz are separate network names.

None of that stops the server working. It is always reachable by IP address, which is the dependable way in: find it in the router's list of connected devices (often called DHCP Clients, Attached Devices, or Device List), where it appears under its Access Point / mDNS Name rather than as an unnamed device, and browse to that address. Giving the server a DHCP reservation, or a static IP, keeps that address from changing — worth doing regardless, since the clients pointed at this server need a stable address.

Router settings vary enormously between makes and models, and we cannot troubleshoot individual router configurations. If getting mDNS working is preferred over using the IP address, we recommend asking an AI assistant such as Claude or ChatGPT — they are genuinely good at this and can give the exact steps for the particular hardware. Tell it the router make and model and what needs to be done.

5. The Status Page

Browsing to the server's address shows a live status page that refreshes once per second without reloading. It is the primary window into what the server is doing.

Field	Meaning
UTC	The current served time in Coordinated Universal Time
Source	The active discipline source: GPS, NTP, or — (none yet)
Status	LOCKED (serving), HOLDOVER (GPS lost but riding the RTC), or NO SYNC
Last sync	How long ago the active source last corrected the clock
Clock temp	The RTC chip temperature, in the unit chosen on the Weather page
BME280 or BME68x	Temperature, barometric pressure and humidity from a fitted BME sensor, in the units chosen on the Weather page. The row is labelled with the part actually detected. An arrow beside the pressure shows the 3-hour trend (double arrow = fast); no arrow means steady. A BME68x also shows its air-quality index. A dash means the sensor has not produced its first reading yet (normal for a moment after start-up); Not Available means a sensor that was detected has stopped answering
SEN66	Temperature, humidity, PM2.5, carbon dioxide and the air quality index from a fitted SEN66. Shown as its own row: with both fitted the two temperatures differ slightly by design, since each sensor measures its own location. A dash means the sensor has not produced its first reading yet (normal for a moment after start-up); Not Available means a sensor that was detected has stopped answering
WiFi	Connected network and signal strength, or AP mode
NTP	The server's own IP address on the local network
mDNS	The <name>.local address for discovery. Whether that name actually resolves depends on the device asking and on the router, so treat the IP address above it as the reliable one

Upstream	The upstream NTP server's hostname and reachability — OK, NO REPLY, or NO DNS — shown when an upstream server is in use
KoD	The last Kiss-o-Death code from the upstream server, with a count since the last restart and its age — shown only if one has been received
Served	How many NTP requests the server has answered since it last restarted, and the last client address. The count is not carried across a reboot
Uptime	Time since the current lock began. It restarts from zero if the time source is lost and re-acquired, so it measures continuous service rather than time since power-on
Heap	Free memory and fragmentation (diagnostic)
Last reset	Why the server last restarted: power-on, software, panic, etc. (diagnostic)
GPS sats	In GPS mode: how many satellites are being used for the position fix, how many are being tracked, and how many are in view. At least four must be in use before a fix is possible
GPS signal	In GPS mode: the average and best satellite signal strength, in dB-Hz. Green is 35 or above, amber 25 to 34, red below 25. The figures rise as the receiver settles into a steady lock, so low readings shortly after start-up are normal
GPS data	In GPS mode: counts of messages from the receiver since the last restart — good, corrupted, with a fix, and without a fix (diagnostic)
GPS restarts	Appears only if the GPS receiver has reported starting up — normally this row is absent, which is the healthy state. If it appears and the count keeps climbing, the receiver is restarting instead of running continuously and will never hold a fix (diagnostic)
GPS−NTP	In GPS mode: the difference between GPS and the upstream NTP server, in milliseconds, while both are live (a dash otherwise). Turns red on a persistent disagreement over one second.

Every counter on this page — Served, the GPS message counts, the GPS restart count, and the KoD count — is kept in memory only and starts again from zero each time the server restarts. None of them is a lifetime total, so a figure that looks low simply means the server has rebooted recently.

When a fault is active, a red warning line appears at the bottom of the box — GPS lock lost, holdover expired, upstream NTP unreachable, low memory, or Internet unreachable — and disappears the moment the condition clears. These are the same conditions that

can send a notification by email or Telegram (Section 9), shown here the instant they occur regardless of whether notifications are enabled.

6. Settings Reference

The web interface is organized into pages, reached from the buttons on the home (status) page: Settings (network and time), Weather (the location, the environmental sensor, and the Internet weather sources), Notifications, and — once the server is on the network — System. This section covers the Settings page; Weather is described in Section 7 and Notifications in Section 9.

These options appear on the setup form and, where they apply at runtime, on the live settings page. Save persists them. Most settings apply immediately and Save simply stores them; Save reboots automatically only when a setting that is applied at boot is changed — WiFi network or password, the Access Point / mDNS Name, any Static IP Address field, Discipline Mode, the NTP Server / Alternate NTP Server / NTP Polling Interval, or GPS Baud. All other settings (holdover, LED brightness and color, everything on the Weather page, and everything on the Notifications page) take effect without a reboot. When Save does reboot, the page waits for the server to come back and then returns to the home status page automatically. If the static IP was changed, the page follows the server to its new address on its own, since that address is known. The one case it cannot follow is a change to the WiFi network or password: the server rejoins and is given a new address by the router that is not known in advance, so reconnect to it manually at infoserver.local (or its new address).

Time Source

Setting	What it does
Discipline Mode	NMEA GPS (default; GPS primary, NTP fallback) or Upstream NTP (no GPS; NTP is the only source)
NTP Server	Primary upstream time server, used as the fallback in GPS mode or the source in NTP mode (up to 64 characters)
Alternate NTP Server	A second server the firmware rotates to if the primary is unreachable; leave blank to disable (up to 64 characters)
NTP Polling Interval (minutes)	How often to re-sync the clock from the NTP server (5 minutes to 4 hours). Does not affect boot or retry timing
NTP Fallback	How long GPS keeps discipline through a dropout before NTP takes over (1 hour to 1 day; default 4 hours)
GPS Baud	Serial speed of the GPS module (this receiver ships at 9600)

LEDs

Setting	What it does
Indicator LEDs	Brightness of the two top status LEDs: Disabled, Low, Medium, or High
Under-light Color	The color of the two bottom LEDs. The color also sets their brightness — a darker shade dims them, black turns them off

System

The System sub-page appears once the server is running on the network. It collects the actions and settings that affect the server itself rather than the time or information services:

Item	What it does
Switch to AP Mode	Restarts the server into setup mode so the WiFi network can be changed. The confirmation names the hotspot to join afterwards
Add to Home Screen	Installs this web interface on a phone or tablet as an app with its own icon, so it opens directly without typing an address. Purely local — nothing is sent anywhere, and it can be removed like any other app
Update Firmware	Checks for and installs new firmware immediately (Section 12)
Firmware URL	Where the server looks for firmware (up to 128 characters). Leave this at the factory setting unless instructed otherwise
Automatic Updates	When on (the default), the server checks weekly, overnight, for new firmware and installs approved updates on its own. Settings are always preserved. Turn it off to install updates manually (Section 12)

Temperature Units and Pressure Units, under Sensor on the Weather page, set what the status page displays for the clock and for any fitted sensor. This affects the status page only — the data published at /info always contains every unit.

7. Information Services

Beyond serving time, the InfoServer can publish read-only information about its environment and location on the local network as simple JSON — so other devices, scripts, and dashboards can read it without any cloud service. This is the "info" in InfoServer. The available information includes readings from the optional on-board environmental sensor, current weather from configured Internet providers, active National Weather Service alerts, and computed sun and moon data. These sources are configured on the Weather page of the web interface, which starts with the shared Location fields (latitude and longitude in decimal degrees) used by every location-based source.

Setting the location

Every location-based source — the weather providers, NWS alerts, and sun and moon — works from a single latitude and longitude, set once at the top of the Weather page. They can be typed directly in decimal degrees (positive north and east, negative south and west), or pick the spot on the map above the fields, which is easier and is the usual way to do it. Tapping the map, or dragging the pin, fills in both fields automatically. Accuracy of a few hundred metres is ample; nothing here is sensitive to more than that.

The map is locked when the page opens, and a Lock Map checkbox sits just below it. While it is ticked the map can be panned and zoomed freely to look around, but neither tapping nor dragging the pin will change the location — this prevents a stray touch from silently moving it while scrolling past on a phone. Untick the box to make a change, then set it as required. The lock is not a saved setting: the map is locked again the next time the page is opened.

The map needs an Internet connection, because the map images come from a public map service rather than from the server. During first-time setup, while connected to the server's own access point and it has no route to the Internet, the map is not shown at all and the coordinates are entered in the fields instead. As with every other setting, a location picked on the map is not stored until Save is pressed.

The /info manifest

Any device on the network can request http://infoserver.local/info (or the server's IP address followed by /info) and receive a manifest: a small JSON document listing the device, its firmware version, and every service currently available. The time service is always listed; each optional information source appears only while it has fresh data to

offer. A client reads the manifest, then fetches whichever sources it cares about from the endpoints named there.

Endpoint	Provides	Appears when
/info	The manifest: device name, firmware version, and the list of services	Always
/info/sensorBME	Temperature, barometric pressure, humidity and the 3-hour pressure trend from a BME280 or BME68x — plus air quality on a BME68x	A BME sensor is fitted and reading
/info/sensorSEN6x	Temperature, humidity, PM1/PM2.5/PM4/PM10, carbon dioxide, VOC and NOx index, and the air quality index, from a SEN66	A SEN66 is fitted and reading
/info/om	Current weather, today's and tomorrow's high/low and precipitation chance, and UV index from OpenMeteo	Enabled, with a location set and fresh data — no API key needed
/info/owm	Current weather from OpenWeatherMap	Enabled, with a key and a location set and fresh data
/info/vc	Current weather from Visual Crossing	Enabled, with a key and a location set and fresh data
/info/nws	Active National Weather Service alerts for the location	Enabled, with a location set and a fresh last check
/info/sun	Today's sunrise, sunset, day length, and the current moon phase	Enabled, with a location set and the clock synced

Each reading is returned with an age in seconds, so a client can tell how current it is. A source with no data — no sensor fitted, no API key, no location, or data gone stale — is simply omitted from the manifest rather than returning an error, so clients degrade cleanly.

A fitted environmental sensor is published the same way, at /info/sensorBME or /info/sensorSEN6x. Section 8 covers the sensors themselves — what may be fitted, and how the air-quality numbers are worked out.

Weather sources

The server can fetch current conditions from up to three free Internet weather providers — OpenMeteo, OpenWeatherMap, and Visual Crossing — and republish them on the LAN at /info/om, /info/owm, and /info/vc. To enable a source, open the Weather page in the web interface, enter the location, turn on that provider's Enable toggle, and — for OpenWeatherMap and Visual Crossing — paste in the provider's free API key. OpenMeteo needs no key at all, which makes it the quickest one to switch on.

- Location — the latitude and longitude in decimal degrees, at the top of the page and shared by every Internet source. In Google Maps, right-click the spot and click the coordinates to copy them.
- Enable toggle — each provider has its own on/off switch (off by default). Turning a source off pauses its fetching and serving without erasing the key below it, so it can be switched back on at any time.
- OpenMeteo — free worldwide service, no API key or account. Its data goes beyond current conditions: /info/om also carries today's and tomorrow's forecast high and low, the chance of precipitation for both days, and today's maximum UV index. (Precipitation probability is a forecast product the model occasionally omits; it is then served as -1, meaning "not provided.")
- OpenWeatherMap — paste a free key from openweathermap.org.
- Visual Crossing — paste a free key from visualcrossing.com.
- Poll Interval (minutes) — how often the server refreshes each provider (5 to 60). The defaults sit comfortably inside each provider's free tier.

A source that is toggled off, is missing its key, or has no location set is absent from the manifest. If a provider's data goes stale — older than an hour, for instance after an Internet outage — that source drops out of the manifest until a fresh reading arrives, so clients never receive outdated weather. Weather is fetched in the background and never delays time service.

NWS weather alerts

The server can also watch for official National Weather Service alerts — warnings, watches, and advisories — covering the configured location, and republish them on the LAN at /info/nws. The NWS service is free, needs no API key, and covers the United States and its territories. Turn it on with the Enable Alerts toggle under NWS Weather Alerts on the Weather page; the Poll interval slider (2 to 15 minutes, default 5) sets how often the server checks.

Alerts are matched to the exact coordinates: the NWS resolves the Location fields at the top of the page to the county and forecast zone, and only alerts whose area actually covers that point are returned — a storm warning three counties away never appears. When more than one alert is active at once, the served data reports the total count and the details of the most serious one (ranked by severity, then urgency, then which ends soonest).

The /info/nws reading includes the number of active alerts, the top alert's event name (for example "Tornado Warning"), its severity both as a word (Minor, Moderate, Severe, Extreme) and as a number from 1 to 4 that simple displays can show directly, its urgency, certainty, and recommended response, the minutes remaining until it ends, and its one-line headline. A count of zero is a confirmed all-clear — the source stays in the manifest with fresh data. Because a stale warning is worse than none, alert data older than 15 minutes drops the source from the manifest until the next successful check; with

the poll interval capped at 15 minutes, a working connection never expires between checks.

Sun and moon

With the Enable Sun & Moon toggle on, the server publishes today's sunrise and sunset, the day length in minutes, and the current moon phase at /info/sun. Unlike the weather sources, nothing is fetched from the Internet: everything is computed on the server itself from the Location and the disciplined clock, using the standard astronomical formulas. That means it works with no Internet connection, has no key, no poll interval, and can never go stale — every request is calculated fresh at that moment, and the age reported is always zero.

Sunrise and sunset are served both as Unix timestamps (which a client device converts to its own local time) and as UTC clock strings. The moon is described three ways: its age in days through the 29.5-day cycle, its illuminated percentage, and its phase name — New Moon, Waxing Crescent, First Quarter, Waxing Gibbous, Full Moon, Waning Gibbous, Last Quarter, or Waning Crescent. At extreme polar latitudes, on days when the sun never rises or never sets, the day length is reported as -1 and the rise and set fields are blanked; the moon data is unaffected. The only requirement is a synced clock, so the source appears in the manifest shortly after boot once a time source has locked.

Worked examples

Everything the server knows is available to anything on the network as plain JSON over ordinary HTTP — no account, no key, no library, and nothing to install. The examples below are real responses, trimmed only where a list repeats or a field list runs long. They are the quickest way to see the shape of the data before writing anything against it.

Start with the manifest, which lists every service currently on offer:

```
http://infoserver.local/info

{
  "device": "InfoServer",
  "fw": "1.8.0",
  "services": {
    "ntp": { "proto": "udp", "port": 123, "ref": "GPS",
             "stratum": 1, "state": "locked" },
    "sensorBME": { "endpoint": "/info/sensorBME", "source": "BME688",
                   "fields": "tempC,tempF,...,iaqCategory,co2Eq,bvocEq",
                   "age_s": 2 },
    "sensorSEN6x": { "endpoint": "/info/sensorSEN6x", "source": "SEN66",
                     "fields": "tempC,tempF,...,aqiCategory,co2Category",
                     "age_s": 1 },
    "om": { "endpoint": "/info/om", "source": "OpenMeteo",
            "fields": "tempC,tempF,...,popPct,pop2Pct",
            "age_s": 240 },
    "sun": { "endpoint": "/info/sun", "source": "SunMoon",
             "fields": "riseEpoch,...,moonIllum,moonPhase",
```

```
            "age_s": 0 }
  }
}
```

Every entry gives the address to fetch, where the reading came from, the exact field names it will contain, and how many seconds old it is. The source names the exact part that was detected, so a client can tell a BME280 from a BME68x without asking. The field lists are long, so they are abbreviated with an ellipsis above; the server sends them in full. Then request whatever is needed. Indoor conditions from a fitted BME sensor:

```
http://infoserver.local/info/sensorBME

{ "source": "BME688", "age_s": 2, "fields": {
    "tempC": 22.41, "tempF": 72.34, "tempK": 295.56,
    "pressPa": 101325.0, "pressHpa": 1013.25, "pressInHg": 29.921,
    "rh": 43.20,
    "trendHpa": -1.35, "trendMin": 180, "trend": "falling",
    "gasOhm": 152340, "iaq": 74.5, "iaqStatic": 71.2,
    "iaqAccuracy": 3, "iaqCategory": "Good",
    "co2Eq": 612.4, "bvocEq": 0.68 } }
```

The last six fields appear only on a BME68x. On a BME280 the reply stops after the pressure trend, and the manifest's field list says so in advance. While a BME68x is still calibrating, iaqCategory reads calibrating instead of naming a band. iaq is Bosch's standard index and iaqStatic its static version; iaqCategory is rated on iaqStatic.

And from a SEN66, if one is fitted:

```
http://infoserver.local/info/sensorSEN6x

{ "source": "SEN66", "age_s": 1, "fields": {
    "tempC": 21.86, "tempF": 71.35, "tempK": 295.01, "rh": 45.90,
    "pm1": 3.2, "pm2p5": 4.8, "pm4": 5.1, "pm10": 5.3,
    "co2": 842, "vocIndex": 118.0, "noxIndex": 1.0,
    "aqi": 27, "aqiDriver": "pm2p5", "aqiCategory": "Good",
    "co2Category": "Fair" } }
```

aqiDriver names the pollutant that set the index, so a high reading can be acted on rather than merely noticed.

Outdoor weather and the two-day forecast:

```
http://infoserver.local/info/om

{ "source": "OpenMeteo", "age_s": 240, "fields": {
    "tempC": 18.30, "tempF": 64.94, "tempK": 291.45,
    "pressPa": 101200.0, "pressHpa": 1012.00, "pressInHg": 29.885,
    "rh": 57.00,
    "windMs": 3.20, "windMph": 7.16, "windKph": 11.52, "windDeg": 245,
    "conditions": "partly cloudy",
    "hiC": 24.1, "hiF": 75.4, "loC": 12.8, "loF": 55.0,
    "hi2C": 26.0, "hi2F": 78.8, "lo2C": 14.2, "lo2F": 57.6,
    "uvMax": 6.4, "popPct": 10, "pop2Pct": 35 } }
```

The remaining addresses work the same way: /info/owm and /info/vc for the other two weather providers, /info/nws for severe-weather alerts, and /info/sun for sunrise, sunset, day length and moon phase.

Three conventions make this easy to build against. Every reading arrives in every unit — Celsius, Fahrenheit and Kelvin; pascals, hectopascals and inches of mercury; meters per second, mph and km/h — so nothing on the client end has to convert anything. The age_s value always states how old the reading is. And a source that is switched off or has gone stale drops out of the manifest and returns a clean "not found" rather than an error or an empty shell, so a client device can always distinguish no severe weather from no alert service: an empty alert list is a confirmed all-clear.

Because it is nothing but ordinary HTTP and JSON, anything can read it — a home-automation system, a Raspberry Pi, a Python script, a spreadsheet, a phone shortcut, or one of our own clocks.

8. Environment Sensors

Three different environment sensors are supported, and fitting one is optional. A BME280 or BME68x mounts directly on the board, or connects with a short ribbon cable, and measures temperature, barometric pressure and humidity, the BME68x adding air quality. A Sensirion SEN66 connects with a short ribbon cable and sits away from the board, measuring particulates, carbon dioxide, temperature, humidity and airborne chemicals. Any one may be fitted on its own, and a SEN66 may be paired with either a BME280 or a BME68x. There is nothing to set up: at power-on the server asks each sensor to identify itself, so no setting can disagree with the hardware.

Pairing a SEN66 with a BME is the most capable combination, because the SEN66 has no barometer: the BME fills that gap, and the server feeds its pressure back to correct the SEN66's carbon-dioxide reading. Their temperature and humidity will not agree exactly, since each is measuring its own spot, so every reading is labelled with the sensor that took it.

Air quality with a BME68x alone

A BME68x has a heated plate whose resistance changes in the presence of airborne chemicals — cooking, solvents, cleaning products, people. It cannot identify any individual gas or measure a concentration; it senses the room's overall chemical load, which Bosch's software turns into an index from 0 to 500. Bosch's calculation is unpublished. What it does is documented: it weighs the plate's resistance against humidity and against a baseline of the cleanest air seen recently, re-learning that baseline over about four days — so read the result as a change, not a measurement. The description and status page use Bosch's static index, the version meant for a device that stays in one place. The sensor also reports its confidence, 0 to 3; below 2 the server publishes calibrating instead of a description, and calibration survives restarts. A carbon-dioxide equivalent is reported too — inferred, not measured — named co2Eq.

The scale runs 0–50 Excellent, 51–100 Good, 101–150 Lightly polluted, 151–200 Moderately polluted, 201–250 Heavily polluted, 251–350 Severely polluted, and 351–500 Extremely polluted.

Air quality with a SEN66 alone

A SEN66 measures real concentrations, so its index is the published US EPA Air Quality Index, worked out from the particulate readings on the EPA's current (2024) breakpoints. Each particle size becomes a sub-index by straight-line interpolation between the two

breakpoints bracketing it. C is the concentration in micrograms per cubic metre, I the index, and (lo)/(hi) the bracketing row:

```
        I(hi) - I(lo)
I  =  ---------------  x  ( C - C(lo) )  +  I(lo)
        C(hi) - C(lo)
```

Index and description	PM2.5	PM10
0–50 Good	0.0–9.0	0–54
51–100 Moderate	9.1–35.4	55–154
101–150 Unhealthy (sensitive)	35.5–55.4	155–254
151–200 Unhealthy	55.5–125.4	255–354
201–300 Very Unhealthy	125.5–225.4	355–424
301–500 Hazardous	225.5 up	425–604

The reported index is the higher of the two sub-indices — the EPA's rule, since air quality is set by whichever pollutant is worst, not by their average — and the server publishes which one it was. One adjustment matters: the EPA index is defined on a 24-hour average, and a live reading would be useless indoors, where one candle would drive it to Unhealthy for a minute. The server therefore applies NowCast, the EPA's method for live sensors, which averages the last twelve hours but weights recent ones more when the air is changing:

```
          max - min                    SUM ( c(i) x w^i )
w  =  1 - -----------      NowCast = ------------------
             max                          SUM ( w^i )
```

Here c(i) is the hourly average i hours ago, i=0 being the newest, and w is never allowed below 0.5.

Carbon dioxide is reported separately and kept out of the index deliberately: indoors it indicates ventilation rather than harm. Below 600 ppm is Excellent, then Good to 800, Fair to 1000, Poor to 1400, Very poor to 2000, Severe above — common practice, not a legal standard. The volatile-compound and nitrogen-oxide indexes are plain 1-to-500 numbers with no description: they track the room's own average over the past day, so 100 means normal for wherever it stands.

Air quality with both fitted

With both fitted, both indexes are calculated and published, each on its own endpoint. Do not expect them to agree — they measure different things by different means and share no common reference, so a BME68x may report Lightly polluted while the SEN66 reports Good, and neither is wrong. Read the SEN66 for how clean the air measurably is, and the BME68x for whether the room's chemical load has changed.

9. Notifications

The InfoServer can send a notification when something affects its ability to keep good time — for example if GPS reception is lost or the upstream Internet time server becomes unreachable. Two delivery channels are available, and either or both can be used: email, sent through a standard mail provider, and Telegram, sent as instant messages from a dedicated free Telegram bot. Both channels share one list of alert events — the events to report are chosen once, and every enabled channel delivers them. Notifications are off by default; enable and configure them on the Notifications page of the web interface.

How alerts behave

Alerts are designed to inform without nagging:

- One message per event. A standing problem — say, GPS staying lost for hours — produces a single message when it begins, not a repeat every few minutes.
- A 60-second settling time. A problem must persist for a full minute before it notifies, and must stay resolved for a full minute before the server considers it cleared. Brief momentary glitches are ignored.
- Grouped messages. If several problems occur close together (within about ten seconds), they are combined into a single message rather than arriving separately.
- Optional recovery notice. Turn on Notify on Recovery to also receive one message when a problem clears. Most faults send one; the one-time events — KoD Received, Device Rebooted, and Time Source Changed — have nothing to clear, so they never do.

Whether or not any notification channel is enabled, any active problem is also shown as a red line at the bottom of the status page the moment it occurs, and disappears when it clears (Section 5).

Setting up email (SMTP)

The server sends mail through a standard outgoing (SMTP) mail server — typically the email provider's. Fill in the SMTP Connection and Envelope fields:

Field	What to enter
Enable Email Alerts	Turn on to allow the server to send mail. When off, nothing is ever sent
SMTP Host	The provider's outgoing mail server, e.g. smtp.gmail.com for Gmail
Security	The connection type, which also sets the port: SSL (465), STARTTLS (587), or None (25). Gmail uses SSL (465)
SMTP Username	The login for the mail server — usually the full email address
SMTP Password	An app-specific password for the account (see below), not the normal login password
From Address	The sender address. For Gmail this must match the username
From Name	An optional display name for the sender (for example InfoServer)
Send To	Where alerts go. Separate multiple recipients with commas

Getting a Gmail App Password

Gmail and Google Workspace will not accept the normal account password from a device. Instead, create a one-off, revocable App Password. This requires 2-Step Verification to be turned on for the account first.

- 1. Turn on 2-Step Verification at myaccount.google.com → Security, if it is not already on.
- 2. Go directly to myaccount.google.com/apppasswords. Google no longer lists this page in the Security menu, so the direct address is the reliable way to reach it.
- 3. If prompted, re-enter the Google password.
- 4. Under "App name," type a memorable label such as InfoServer, and click Create.
- 5. Google shows a 16-character password in four groups of four. Enter it into the SMTP Password field without the spaces.

Use the normal Gmail address for both the SMTP Username and the From Address. If the App Passwords page says the option is not available, 2-Step Verification is not fully enabled yet, or an administrator has disabled app passwords on a Workspace account.

Setting up Telegram

Telegram is a free messaging app (telegram.org) for phones, tablets, and computers. As an alert channel it has some real advantages over email: messages arrive instantly as notifications on every device running Telegram, nothing lands in a spam folder, and there is no mail server or password to configure. Alerts are delivered by a dedicated Telegram bot — a private automated account that is created once, in about two minutes, using the steps in the next section. The server needs just two pieces of information, both on the Notifications page:

Field	What to enter
Enable Telegram Alerts	Turn on to allow the server to send Telegram messages. When off, nothing is ever sent
Bot Token	The secret token BotFather issues when the bot is created — the bot's password. See the steps below
Chat ID	The numeric Telegram ID of the person (or group) that receives the alerts. Not a phone number — press Get Chat ID and the server fills this in automatically

Two buttons under the fields complete the setup: Get Chat ID looks up and fills in the Chat ID automatically, and Send Test Telegram confirms the whole channel end to end.

Getting the Bot Token and Chat ID, step by step

Bots are created by talking to BotFather — Telegram's own official bot-creation assistant. The whole process happens inside the Telegram app; nothing is installed and nothing is paid for.

- 1. If Telegram is not already in use, install it from telegram.org (or the device's app store) and sign in with a phone number.
- 2. In Telegram's search box, type BotFather and open the result named BotFather with the blue verified check mark (its full handle is @BotFather). Tap or click Start.
- 3. Send it the message /newbot. BotFather first asks for a display name for the bot — this is just a label, so type anything, for example InfoServer Alerts.
- 4. It then asks for a username for the bot. This one must be unique across all of Telegram and must end in the letters bot — for example SmithInfoServer_bot. If the first choice is taken, try a more distinctive one.
- 5. BotFather replies "Done!" with a long access token that looks like 8123456789:AAHrX9k2mQ... — this token is the Bot Token. Tap it in the chat to copy it.
- 6. Open the InfoServer's Notifications page and paste the token into the Bot Token field.
- 7. Back in Telegram, open a chat with the new bot — BotFather's "Done!" message includes a t.me link to it; tap that link, then tap Start (or simply send the bot any message, such as "hello"). Do not skip this: the server can only discover the Chat ID after the bot has received at least one message from that account.
- 8. On the Notifications page, press Get Chat ID. The server asks Telegram who last messaged the bot and fills in the Chat ID field, confirming with the account name — for example "Chat ID 123456789 (Sam) — saved." The ID is stored immediately; there is no need to press Save for it.
- 9. Turn on Enable Telegram Alerts and press Send Test Telegram. The test message should appear in the Telegram chat within a few seconds.

Keep the token private — anyone who has it can send messages as the bot. If it ever leaks, send /revoke to BotFather to invalidate it and issue a fresh one, then paste the new token into the Bot Token field.

To deliver alerts to a family or team group instead of one person, add the bot to the group, send any message in the group, and press Get Chat ID — group IDs are negative numbers, which is normal. And if Get Chat ID ever reports that no messages were found, simply send the bot (or group) another message and press the button again: Telegram only holds a bot's undelivered updates for about 24 hours.

Choosing which events send notifications

Each event type has its own on/off toggle; all are on by default. Turn off any that are not wanted. The event list is shared by both channels — each event left on is delivered by email, Telegram, or both, according to which channels are enabled.

Event	Alerts when
GPS Lock Lost	GPS drops a valid fix and the server falls back to the DS3231 holdover
Holdover Expired	The holdover window runs out and the server switches to the Internet NTP fallback
NTP Unreachable	The upstream NTP server cannot be resolved or stops replying
Internet Down	WiFi is up but nothing beyond the router responds. The server probes the Internet every 5 minutes and the fault trips after two consecutive misses. The outage itself usually blocks the onset message — with Notify on Recovery on, the "Internet restored" message is the one that reliably arrives
KoD Received	The upstream NTP server sends a Kiss-o-Death asking the server to back off; the message includes the code, with a plain-English explanation for the common ones (RATE, DENY, RSTR)
Device Rebooted	The server restarts — one message, including the reason for the restart
Low Heap / Health	Free memory falls to a level that may affect stability
Time Source Changed	The served time source flips between GPS and NTP in either direction — GPS to NTP when GPS is lost past the holdover window, and NTP back to GPS when GPS returns. One message per switch
Weather Alert	The NWS issues a new alert for the configured location at or above the Minimum Severity set below it. Requires NWS Weather Alerts to be enabled on the Weather page. One message per new alert, with its headline and severity

The Weather Alert event has one extra setting, Minimum Severity: Any, Moderate, Severe, or Extreme. Only alerts at or above the chosen level are sent — set it to Severe, for instance, to hear about severe thunderstorm and tornado warnings but not routine advisories. The choice affects notifications only; the /info/nws data always carries every active alert regardless. With Notify on Recovery on, one further message arrives when the last qualifying alert has cleared.

About time-source switching

In NMEA GPS mode the server normally disciplines its clock from GPS. When GPS reception is lost it does not switch away immediately: it rides the battery-backed DS3231 holdover — still GPS-anchored and still stratum 1 — for the configured NTP Fallback window (Section 6). Only if GPS has not returned by the end of that window does the served time source actually switch to the upstream Internet NTP server. When GPS later reacquires a fix, the source switches back to GPS. Each of these switches is a change in which source the devices are being served from.

The alert events cover this in two complementary ways, so the level of detail can be chosen:

- Time Source Changed fires once on every actual switch, in both directions — a single "GPS to NTP" or "NTP to GPS" message. This is the simplest way to be notified only when the source the clients rely on has really changed.
- GPS Lock Lost and Holdover Expired break the GPS-to-NTP direction into its two stages: GPS Lock Lost the moment the fix drops (the server is now on holdover), and Holdover Expired when the window runs out and NTP actually takes over. The return to GPS is reported by Time Source Changed, and — if Notify on Recovery is on — also as "GPS lock restored" / "Holdover recovered".

Because these overlap, a single GPS-to-NTP switch can, with everything enabled, produce a Time Source Changed message and (about a minute later, after its settling time) a Holdover Expired message. For only one, leave Time Source Changed on and turn Holdover Expired off, or vice versa. In pure Upstream NTP mode there is no GPS, so no source switching occurs and none of these fire.

Testing

Each channel has its own test button. After filling in the email connection and envelope fields, use Send Test Email; after setting up Telegram, use Send Test Telegram. In both cases the server sends a test message and reports the result on the page — either confirmation that it was sent, or the specific error that came back: a wrong password or host for email, an invalid token ("Unauthorized") or wrong chat ID ("chat not found") for Telegram. This is the quickest way to confirm the settings before relying on alerts. The tests work even while a channel's master Enable is off, as long as its fields are filled in.

10. Indicator LEDs and Under-light

The server has four LEDs in one chain: two status indicators on top and two under-lights on the bottom.

Status LEDs (top)

The two top LEDs report what the server is doing. The STATUS LED pulses one short heartbeat per second; its color shows the active state — green for GPS, blue for NTP, amber for GPS holdover. The ACTIVITY LED flashes that same color as each time request is served, and flashes purple when an information request is served.

State	STATUS LED	ACTIVITY LED
Starting up	● Red — solid until the first time source locks	Off
GPS locked	● Green — one short heartbeat pulse per second	● Green flash on each NTP time request served
NTP locked	● Blue — one short heartbeat pulse per second	● Blue flash on each NTP time request served
GPS holdover	● Amber — one heartbeat pulse per second	● Amber flash on each NTP time request served
Information served	Heartbeat continues (source color)	● Purple flash on each /info request served
No sync	Off	Off
AP setup mode	● Red — alternating, ½ second	● Red — alternating, ½ second
Firmware update	● Solid red	● Solid red
IP-address readout	Blinks the IP out octet by octet (Section 11)	Alternates green/red with STATUS to mark each digit

At power-on the STATUS LED is solid red until the server locks to its first time source, at which point it switches to the green, blue, or amber heartbeat above. The ACTIVITY LED gives a brief flash on each request answered: an NTP time request flashes in the same color the STATUS LED is showing (green on GPS, blue on NTP, amber in holdover), while an information-service request (the /info endpoints, Section 7) flashes purple — so time traffic can be told from info traffic at a glance. The Indicator LEDs setting (Section 6) dims or disables the normal GPS, NTP, holdover, and activity indications; the attention states — AP setup mode, firmware update, and IP-address readout — always show at full brightness regardless of that setting.

Under-light (bottom)

The two bottom LEDs are decorative under-lighting set to a single constant color by the Under-light Color picker (blue by default). Because the color carries its own brightness, picking a darker shade dims the under-light and choosing black turns it off entirely. The under-light is independent of the Indicator LEDs brightness setting.

11. The SW1 Button

Action	Result
Double-click	Blink out the server's IP address on the LEDs, one octet at a time
Hold 5 seconds	Restart into AP (setup) mode temporarily, to reconfigure WiFi or settings — the full setup form appears, and Start (or a power cycle) returns the server to normal operation
Hold 10 seconds	Check for and install a software update over WiFi (see Section 12)
Hold during power-on	Factory reset — wipe all settings and reboot into AP setup mode. This is the only action that clears the configuration; software updates never do

The two timed holds share one button, so the duration matters. Release at about five seconds for AP mode; keep holding to about ten seconds for the update check. Watch the LEDs: both go solid red when the update begins. A hold during power-on (before the LEDs come up) is the factory reset.

Reading the IP blink

After a double-click, the IP address is signalled octet by octet. Each digit is shown as that many blinks (a zero is shown as a single long flash), with the two LEDs alternating to mark digit boundaries, and a longer pause between octets. For example, 192.168.4.75 blinks as 1-9-2, then 1-6-8, then 4, then 7-5. For most users the status page is an easier way to read the address; the blink readout is there for when no screen is handy.

12. Over the Air Updates

New firmware installs over WiFi, with no cable and nothing to download to a computer. Updates can be triggered by hand at any time, and the server can also install them on its own overnight. Both routes use the same mechanism and produce the same result.

Settings are never lost

An update preserves everything: the WiFi credentials, time-source choice, location, weather configuration, notification setup, and LED preferences all carry across untouched, no matter how large the version change. The server never returns to setup mode because of an update. The only thing that clears the settings is a deliberate factory reset — holding SW1 while applying power (Section 11).

Updating on demand

Either hold SW1 for ten seconds while the server is running, or press Update Firmware on the System sub-page. The button is usually easier, and it reports the outcome on the page. (A shorter, five-second hold switches to AP mode instead — see Section 11 — so keep holding past five seconds for an update.) Both status LEDs go solid red while the update runs.

The server fetches a small version file from the configured Firmware URL and compares it with the running version. If they match, it reports that it is already current and resumes normal operation. If a newer version is available, the page shows "updating to v<new version>…", and the server downloads and installs it, then reboots into the new firmware. The whole process is automatic once triggered; do not remove power while both LEDs are solid red.

The Firmware URL is set on the System sub-page and points at the location hosting the version file and firmware image. Leave it at the factory setting unless instructed otherwise.

Automatic updates

Automatic Updates, on the System sub-page, is on by default. With it on the server checks for new firmware about once a week, in the small hours, and installs an update if one is waiting. No action is required, and there is no schedule to configure.

The check runs at a fixed minute between 2:30 and 3:30 AM local time. The exact minute is derived from the server's own hardware address, so it does not change from

week to week, and two units on the same network will not check at the same moment. Local time comes from the Longitude on the Weather page; if no location has been set, the server uses UTC instead. The first check happens on the night after a restart, and roughly every seven days after that.

An automatic run behaves exactly like pressing Update Firmware: the status LEDs go solid red, the firmware installs, and the server reboots. Because it happens overnight, in practice the new version is simply found running the next morning. The service is unavailable for the half-minute or so that the download takes — this is why the check is scheduled overnight rather than during the day.

Not every release installs automatically. Each one is published with a marker naming the release approved for unattended installation; if the waiting release is not the approved one, the automatic check finds nothing to do and leaves the server exactly as it was. This is deliberate — it means an update that warrants a look at the release notes first will wait for the button to be pressed, and the safe outcome (do nothing) is what happens by default. The Update Firmware button ignores the marker entirely and always installs whatever is newest.

If the server cannot reach the network at its appointed minute it retries a few times, a minute apart, and then leaves the matter until the following week. A failed automatic check is not reported and has no effect on timekeeping; the next one simply tries again.

Turning Automatic Updates off stops the weekly check completely. Updating by hand continues to work exactly as described above, so this is the setting to use to decide manually when each version goes on.

13. Specifications

Item	Specification
Controller	ESP32-S3 (240 MHz dual-core, WiFi)
Time source	DS3231 TCXO RTC (±2 ppm), battery-backed
Reference (GPS mode)	u-blox GPS, NMEA + 1PPS — stratum 1
Reference (NTP mode)	Upstream NTP server — stratum 3
LAN accuracy (GPS)	Approximately ±1 ms served; clients typically see ±15 ms end to end over WiFi
Holdover drift	About 7 ms per hour on the DS3231 after GPS loss
Protocol	NTP v4 (RFC 5905) over UDP port 123
Environmental sensor	Optional. BME280 or BME68x — temperature, barometric pressure, humidity (BME68x adds air quality). SEN66 — particulates, CO_2, temperature, humidity, VOC and NOx. Either or both
Info services	JSON /info manifest publishing on-board sensor readings with pressure trend, OpenMeteo / OpenWeatherMap / Visual Crossing weather, NWS weather alerts, and computed sun & moon data
Notifications	Email (SMTP — SSL, STARTTLS, or unencrypted) and Telegram bot messages on time-source faults
Network	2.4 GHz WiFi; DHCP or static IP; mDNS discovery
Software updates	Over-the-air, on demand or automatically overnight; settings always preserved
Power	5 V USB-C

14. Troubleshooting

Symptom	What to check
Source stays at — / NO SYNC	In GPS mode, give the unit a clear view of the sky and allow several minutes for a first fix. Confirm WiFi is connected so the NTP fallback can also work.
Never gets a GPS fix	Move the unit to a window or outdoors; cold starts under cover can take a long time or never complete. The status page Source will read NTP while GPS is unacquired. Check GPS sats — four must be in use before a fix is possible — and GPS signal, which should climb well above 25 dB-Hz with a clear view of the sky.
A GPS restarts row appears and keeps climbing	This row is hidden unless the receiver reports restarting, so its presence is itself the symptom: the receiver is restarting instead of running continuously, and cannot stay locked long enough to complete a first fix. Make sure the unit is running the current firmware.
GPS signal stays low	Weak readings are normal indoors and immediately after start-up, and they climb as the receiver settles into a continuous lock. Persistently low figures are usually a symptom rather than the cause — a signal estimate only settles under continuous tracking. Check whether a GPS restarts row has appeared, which would point to the receiver restarting rather than to poor reception.
Status shows HOLDOVER	GPS lost its fix but the server is still serving GPS-derived time from the RTC. If it persists, improve the unit's view of the sky; after the NTP Fallback window the server switches to NTP.
GPS−NTP shown in red	GPS and the upstream NTP server disagree by over a second. Suspect a GPS fault, poor reception, or a bad upstream server; try an alternate NTP server.
Clients can't reach the server	Confirm the server's IP on the status page and that clients point at it. A DHCP reservation or static IP keeps the address stable.

Can't find the server on the network	Look up its IP address in the router's list of connected devices — it appears under its Access Point / mDNS Name — and browse to that address. Do not rely on infoserver.local here: if the name is what is failing, it cannot also be the test. If the server is absent from that list, WiFi setup did not succeed, and the unit falls back to AP mode — rejoin the InfoServer network and reconfigure.
infoserver.local does not resolve	Expected on some setups — mDNS is not supported by every device, and many routers filter the multicast it depends on (see Section 4). Use the IP address from the router's device list; everything works identically that way. To fix the name resolution itself, ask an AI assistant such as Claude or ChatGPT for the steps for the specific router make and model in use — they handle this well, and router settings differ too much between models for us to advise on individually.
A weather source is missing from /info	Confirm that provider's API key and the latitude/longitude are set on the Weather page, and that the server has Internet access. Data over an hour old is dropped until it refreshes.
/info/nws is missing from the manifest	Confirm NWS Weather Alerts is enabled and a latitude/longitude is set on the Weather page, and that the server has Internet access. The NWS covers US locations only, and alert data over 15 minutes old is dropped until the next successful check.
Internet unreachable shown in red	WiFi is working but nothing beyond the router responds — usually the modem or the provider. Check the modem and the router's Internet (WAN) light. The warning clears itself a few minutes after the connection returns; with Notify on Recovery on, an "Internet restored" message confirms it. The probe makes a TCP connection to a public DNS server (port 53), so a firewall blocking outbound DNS can cause a false alarm.
Test email fails	Check the SMTP host, Security/port, username, and password. For Gmail use a 16-character App Password (Section 9), not the account password, and make sure From matches the username.
No alerts arrive	Confirm the channel's Enable toggle (email or Telegram) and the relevant event toggle are on, and that its test button succeeds. Remember a problem must persist about a minute before it notifies.
Test Telegram fails	"Unauthorized" means the Bot Token is wrong or was revoked — re-copy it from BotFather. "Chat not found" means the Chat ID is wrong — run Get Chat ID again (Section 9).

Get Chat ID finds no messages	The bot has not received a message recently — Telegram holds a bot's updates for only about a day. Send the bot any message in Telegram, then press Get Chat ID again.
Both LEDs alternate fast and won't stop	The unit is not provisioned (authenticity record missing). It must be provisioned before the firmware will run.
Forgotten settings / start over	Hold SW1 while applying power to factory reset into AP mode.

Notices

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Power the server only from a standard 5 V DC USB-C source. The InfoServer contains no user-serviceable parts.

InfoServer — Designed by Mitchell Feig